

WINNEMUCCA, NV, USA

WMO: 725830

Lat: 40.902N

Lon: 117.808W

Elev: 4296

StdP: 12.55

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24128

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.3	9.3	-3.9	5.2	13.9	1.4	6.9	21.4	26.5	44.4	23.0	43.2	3.3	160	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.3	98.4	61.7	95.8	60.5	93.3	59.5	64.6	92.5	62.5	89.8	60.9	87.9	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	75.1	71.0	51.6	66.4	70.2	49.1	60.5	69.7	31.8	92.6	30.1	89.8	28.9	87.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.1	17.1	-5.0	103.2	7.2	3.2	-10.2	105.5	-14.5	107.4	-18.5	109.2	-23.8	111.6		
(5)			-5.8	66.7	7.1	4.1	-11.0	69.6	-15.2	72.0	-19.2	74.3	-24.4	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	31.6	35.6	41.4	46.4	55.0	64.9	74.0	70.7	61.3	48.3	38.1	30.0		
	DBStd	16.45	8.85	7.60	7.12	7.33	7.73	7.63	5.28	5.11	7.39	7.54	8.33	9.43		
	HDD50	2554	570	404	275	153	38	4	0	0	9	120	361	620		
	HDD65	6136	1035	823	730	558	320	92	4	12	153	519	806	1084		
	CDD50	2505	0	1	10	45	193	451	745	643	347	66	4	0		
	CDD65	614	0	0	0	0	10	89	284	190	40	1	0	0		
	CDH74	14432	0	0	2	61	567	2352	5446	4230	1628	146	0	0		
	CDH80	7620	0	0	0	8	177	1131	3223	2352	699	30	0	0		
	WSAvg	7.3	6.7	7.6	8.0	8.2	7.9	8.0	7.4	7.0	6.6	6.4	6.6	6.8		
	PrecAvg	8.2	0.9	0.7	0.8	0.8	0.9	0.7	0.2	0.3	0.4	0.6	0.9	0.9		
(15) Precipitation	PrecMax	15.6	2.0	2.2	1.6	2.9	3.6	2.6	1.7	1.7	2.8	1.7	2.5	3.7		
	PrecMin	4.1	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	2.3	0.6	0.4	0.4	0.6	0.8	0.7	0.3	0.4	0.5	0.5	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	63.0	72.2	80.3	90.2	98.2	102.9	99.8	94.8	84.0	69.5	56.8		
		MCWB	45.2	46.0	48.4	52.8	59.7	61.2	64.8	62.5	60.4	55.2	48.6	43.6		
	2%	DB	52.6	58.0	67.1	75.0	84.7	93.8	99.7	96.8	90.5	78.0	64.7	52.0		
		MCWB	41.8	44.1	46.6	50.9	55.9	59.3	62.0	61.1	57.7	52.3	46.8	42.7		
	5%	DB	48.8	53.4	62.9	70.2	80.8	90.4	97.3	94.5	87.6	73.6	59.7	48.4		
		MCWB	40.1	41.8	45.0	48.5	54.1	57.9	61.2	60.0	56.4	50.0	44.7	40.3		
	10%	DB	44.8	49.5	58.6	65.2	75.8	87.0	94.7	92.0	84.2	69.0	54.5	44.9		
		MCWB	38.2	40.1	43.1	46.6	52.5	56.6	60.0	58.6	55.2	48.1	42.5	38.2		
	0.4%	WB	47.4	47.9	50.2	54.6	61.7	64.6	70.6	66.9	63.5	56.4	50.6	46.4		
		MCDB	54.3	59.4	67.1	77.2	87.6	94.2	99.9	96.3	86.8	79.5	65.2	53.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.5	45.1	47.9	51.7	57.5	61.5	65.4	63.9	60.5	53.7	48.0	43.7		
		MCDB	50.5	55.8	64.2	71.6	79.3	89.8	93.7	91.9	83.9	74.1	62.0	50.0		
	5%	WB	41.1	42.9	46.0	49.6	55.4	59.3	63.1	61.7	58.6	51.4	45.7	41.2		
		MCDB	47.2	52.2	60.8	67.3	76.0	85.5	90.5	88.8	81.9	69.5	58.1	47.6		
	10%	WB	38.7	40.7	44.0	47.5	53.3	57.4	61.5	59.9	56.2	49.2	43.3	38.4		
		MCDB	44.4	48.7	57.2	63.4	72.7	83.3	88.9	86.7	80.5	66.6	53.7	44.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	22.7	27.6	28.9	31.5	35.8	39.3	40.5	39.6	35.6	28.1	22.0		
		MCDBR	26.8	29.0	36.3	40.5	41.0	42.4	42.8	44.4	45.7	44.4	36.8	24.7		
		MCWBR	17.1	17.2	19.2	20.3	18.2	17.6	17.9	19.8	21.3	23.8	21.8	16.6		
		MCDBR	22.1	26.6	32.5	36.3	36.8	39.4	38.9	40.8	40.6	39.9	33.1	21.3		
	5% WB	MCWBR	14.9	16.1	17.8	18.4	16.8	17.6	18.1	19.7	19.4	21.8	20.0	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.256	0.270	0.296	0.312	0.308	0.318	0.312	0.292	0.268	0.253	0.247		
	taud		2.620	2.626	2.586	2.488	2.469	2.514	2.495	2.497	2.553	2.628	2.630	2.622		
	Ebn at Noon		299	312	315	311	305	305	301	301	303	302	295	292		
	Edh at Noon		21	24	28	33	35	33	33	32	29	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		653	972	1386	1816	2173	2535	2499	2215	1799	1246	789	540		
	RadStd		105	115	113	129	184	131	109	106	62	80	55	88		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	+169	+112

Nomenclature: See separate page